Garros Gong, Ph.D.

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Summary

I am an empirical operations management researcher whose work examines how organizations and markets adapt under uncertainty. My research studies how information frictions, incentives, risk exposure, and operational constraints shape decision quality across supply chains, emergency response, prediction markets, and AI-enabled decision-support settings. My senior management experience in large financial organizations strengthens my industry connections and helps me bring applied analytics, finance, and managerial decision-making into the classroom.

Research interests: Sustainable operations management; Supply-chain resilience; Organizational and market adaptation; AI-enabled decision support.

Education

University of Waterloo, Waterloo, Canada **Ph.D. in Mgmt. Sciences (OR)** *Oct 2025*

Dissertation: Data-Driven Decision-Making Under Uncertainty: An Empirical Study of U.S. Wildfire Management.

Research Award: Finalist, Canadian Operational Research Society (CORS) Student Paper Competition (2026).

Ivey Business School, London, Canada **M.Sc. in Mgmt. Sciences** *May 2018*

University of Victoria, Victoria, Canada **B.S. in Fin. Math. and Econ.** *Jun 2016*

Research Award: Jamie Cassels Undergraduate Research Awards (2014).

Peer-Reviewed Publications

- **Sustainable Wildfire Management Meets Social Media: How Virtual Interaction Affects Wildfire Response Costs.**

  Garros Gong, S. Dimitrov, M. R. Bartolacci | Production and Operations Management (2026)

  This paper examines how public visibility and digitally mediated signals reshape resource allocation in emergency response systems. It introduces the Visibility Efficiency Paradox, showing how urgency signals can improve responsiveness while weakening cost efficiency when resource use approaches saturation.

- **Digital strategies in wildfire management: Social media analytics and Web 3.0 integration.**

  Garros Gong, S. Dimitrov, M. R. Bartolacci | Discover Sustainability (2024)

  This paper develops a wildfire decision-support framework using social media analytics for monitoring, prediction, and response, with attention to transparency and system responsiveness.

Conference Presentation

- *Digital Strategies in Wildfire Management: The Advantage of Applying Social Media Analytics and Web 3.0 Integration.* 2024 INFORMS Telecommunications and Network Analytics Conference, Dallas, TX.

Selected Research

- **Benchmarking Supply Chain Resilience: An Exposure-Conditioned Decision Framework.**

  Status: Major Revision | Omega

  This paper develops an exposure-conditioned benchmark for supply-chain resilience using supplier-customer stress and firm operating outcomes. It embeds an auditable resilience measure in a layered decision-support framework with interpretable nonlinear risk diagnostics, validating the framework out of sample against future downside targets.

- **Mechanism Uncertainty in Firm Adaptation to Supply-Chain Shocks: A Causal-Atlas Approach.**

  Solo-authored | Under Peer Review | Management Science

  This paper studies firm adaptation to supply-chain shocks when multiple recovery mechanisms are plausible. It develops a mechanism-atlas approach for evaluating theory-specified mechanism claims across admissible quasi-causal designs, showing that internal buffers and focused product scope provide more stable recovery evidence than visible customer-network reconfiguration under tariff-induced shocks.

- **When Text Helps: AI Text Signals for Calibration-Sensitive Tail-Risk Decisions in Prediction Markets.**

  Status: Resubmitted | Decision Sciences Journal

  This paper studies when AI-extracted public text improves calibration-sensitive downside-risk assessment beyond price-based benchmarks in prediction markets. Using Polymarket event-day data, it positions AI text signals as an auditable measurement layer for risk-sensitive decision support, with value that depends on price-state uncertainty.

- **When Do Budgetary Shocks Backfire? Temporal Decoupling of Fiscal Signals under Climate Risk.**

  Status: Major Revision | Journal of Environmental Management

  This paper documents a preparedness funding paradox in U.S. wildfire management. Using panel fixed-effects and time-series models on 2012–2021 geographic-area data, it shows that positive federal budget shocks reduce preparedness mismatch within the same fiscal year but reverse or attenuate over subsequent budget cycles, while more stable multi-year budget paths better predict preparedness alignment under climate risk.

Research Pipeline

- **Post-Shock Supply-Chain Adaptation.** This research studies when firms should reconfigure supplier-customer networks after supply-chain shocks, and when reconfiguration disrupts relationship-specific efficiency. It extends my resilience-benchmarking and causal-atlas work by linking pre-shock exposure, adaptation choices, and later operating recovery.
- **Climate Shocks and Service-Operations Resilience.** This research uses street-camera video and computer vision to study how climate shocks affect local commercial activity, service recovery, and operating continuity. It extends my sustainable-operations agenda from emergency response systems to observable service activity in affected communities.

Teaching Experience

Centennial College, Toronto, Canada — *Adjunct Professor* *2022–2023*

Financial Analytics: student evaluation 4.25/5.00.

Prepared to teach: Operations Management; Financial Analytics; Business Analytics; AI in Business Decision-Making; Management Science.

Professional Experience

CIBC Mellon, Toronto, Canada — *Director, Profitability and Pricing* *2025–2026*

Led profitability, pricing analytics, and AI transformation initiatives in the finance division, linking cost drivers, client margins, and capital-allocation logic.

Scotiabank, Toronto, Canada — *Manager, Investment Strategist* *2023–2025*

Developed macroeconomic, machine-learning, and factor-model analytics to support portfolio guidance and investment strategy across Canada and Latin America.

Central 1 Credit Union, Toronto, Canada — *Credit and Financial Analyst* *2022–2023*

Supported treasury portfolio analysis, liquidity planning, and risk monitoring.

Marret Asset Management, Toronto, Canada — *Associate, Fixed Income Research* *2021–2022*

Built fixed-income research automation, market-data pipelines, and trading-signal analytics to support investment research and financial operations.

Vinzan International Inc., Toronto, Canada — *Financial and Corporate Analyst* *2018–2020*

Built valuation and pro forma models for corporate finance decisions.

Selected Media Contributions

- Yahoo News (2023): "Increasing Monopoly Power Poses a Threat to Canada's Post-Pandemic Economic Recovery."
- The Conversation (2023): "Canada's New Tech Talent Strategy Aims to Attract Workers from Around the World."
- Toronto Star (2023): "For Gig Economy Millennials, Retirement is Not That Far Off. Innovation Will Be Key to Making It Work."

References and Dissertation Committee

- **External Examiner:** Michael R. Bartolacci (Penn State Berks). mrb24@psu.edu.
- **Committee Member:** Jen Beverly (Univ. of Alberta). jbeverly@ualberta.ca.
- **Supervisor:** Stanko Dimitrov (Univ. of Waterloo). sdimitro@uwaterloo.ca.
- **Internal Members:** Jangho Yang (Univ. of Waterloo). jj34yang@uwaterloo.ca; Selcuk Onay (Univ. of Waterloo). sonay@uwaterloo.ca.
- **Internal–External Member:** Juan Moreno-Cruz. juan.moreno-cruz@uwaterloo.ca.